

Contextual Privacy Pipelines for Frictional Shared Smart Environments

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Abstract

Smart environments increasingly rely on embedded sensors for tasks ranging from safety monitoring to automation. However, privacy management is uniquely difficult in shared spaces because sensing is ambient, spatial, and multi-party: the individuals captured often cannot negotiate with space operators or express context-dependent expectations. We call these settings *frictional shared smart environments*. We argue that privacy mediation here should dynamically synthesize preprocessing pipelines under contextual-integrity (CI) constraints rather than relying on fixed access control or developer-authored policies. We propose *privMediator*, a trusted mediator based on a typed operator-effect model where operators declare their types, utility capabilities, and residual disclosure effects. Given an application request, *privMediator* synthesizes feasible transformations and selects a utility-compatible representation that minimizes context-specific residual disclosure under a configured CI rule library.

We evaluate *privMediator* across 33 scenarios spanning four smart environment tasks (presence, fall, activity, and sound monitoring). Under fixed application contracts, *privMediator* retained 96.5% output utility and made zero releases that violated the configured CI rules, compared with 15 CI-rule violations for raw streams or developer-specified manual pipelines and 11 for an LLM baseline. Under flexible application contracts, *privMediator* retained 91.2% output utility across 21 released scenarios while eliminating CI-rule violations. In a Prolific survey (N=82, 1,476 ratings), *privMediator*'s selected representations received higher average appropriateness ratings than raw media and outputs from developer specified manual pipelines; however, the main advantage is policy level selectivity, since *privMediator* also refuses release in contexts where no CI-feasible output exists. Ultimately, *privMediator* operationalizes contextual integrity at runtime, reducing reliance on tedious human negotiation and manual pipeline authoring while making refusal, utility, and disclosure tradeoffs explicit under deployment-specific rules.

Keywords

contextual integrity, smart environments, data minimization

1 Introduction

Privacy management for software services has become increasingly sophisticated. Modern systems use access-control policies [5, 6], consent flows [20], privacy dashboards [17], and privacy labels [19]

to help users understand and control how applications collect and use data. These mechanisms work best when data collection is tied to a user-facing interface such as an account, browser, phone, app session, or personal device. Smart spaces challenge this model. Cameras, microphones, motion sensors, smart locks, and environmental sensors collect data because people are present in a physical environment, not because each person has intentionally interacted with a particular application or control surface.

This mismatch is especially important in what we call *frictional shared smart environments*: sensor instrumented spaces where privacy decisions would ideally be negotiated among multiple stakeholders, but such negotiation is difficult to perform in practice. The individuals captured by sensors may have limited ability to negotiate with the party that configured the space, such as a building operator, employer, or landlord. They may also have limited ability to negotiate with one another, even though a single sensing decision may simultaneously capture regular occupants, employees, visitors, or bystanders. Existing mechanisms only partially address this setting. Web and mobile permissions assume a user facing control surface; notice-and-choice assumes people can observe, understand, and act on notices; privacy labels and IoT transparency mechanisms can describe sensing practices but do not determine what technical representation should be released [19]; and smart home negotiation mechanisms often rely on repeated relationships or direct interaction between household members [53]. These assumptions weaken in spaces with transient occupants, bystanders, changing purposes, and institutional operators. We focus on infrastructure and environment-integrated sensors controlled by a space operator, rather than arbitrary personal devices carried by visitors.

We propose *privMediator*, a privacy mediator for frictional shared smart environments. Conceptually, *privMediator* is a trusted sensing hub implemented as a virtualized service isolated on a secure platform (e.g., via hardware-enforced isolation) for a particular space. It sits between infrastructure sensors and downstream applications, receives application requests and contextual information, and decides at runtime what transformed representation of a sensor stream should be released. Our stance is to prioritize application utility subject to privacy constraints: the mediator first searches for preprocessing pipelines that can satisfy the application's utility contract, then selects a data-minimized output representation that satisfies the configured CI constraints. When no representation can satisfy both the utility contract and the active configured constraints, *privMediator* returns a rejected application request, which a deployment may later route to consent, human review, emergency escalation, or degraded-task modes.

This decision is difficult because an appropriate preprocessing choice is context-dependent. An occupancy application might need a count in one context, coarse presence in another, and no access in a third. A security application might

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Proceedings on Privacy Enhancing Technologies YYYY(X), 1–22

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<https://doi.org/XXXXXXX.XXXXXXX>



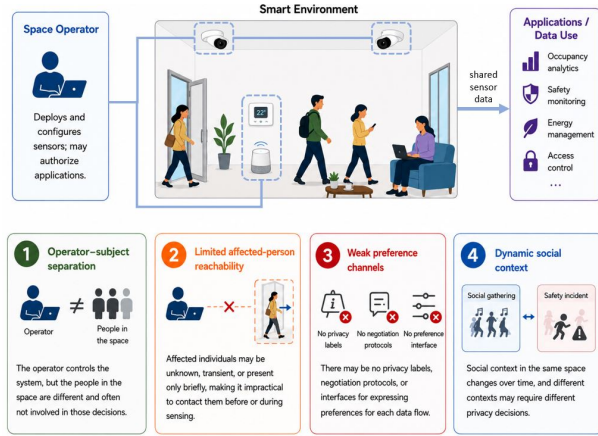


Figure 1: Frictional shared smart environments combine operator-subject separation, limited affected-person reachability, weak preference channels, and dynamic social context. AI-use disclosure: this figure was generated with assistance from ChatGPT and manually edited by the authors.

require detecting visitors at an entryway, while continuous identifiable video may be unnecessary or inappropriate. An audio safety application might need a sound event label (e.g. glass breaking) while speech content should remain inaccessible. Existing enforcement mechanisms often assume fixed access policies or developer-specified pipelines, such as allowing a stream, denying a stream, or applying a predetermined degradation level. *privMediator* instead treats privacy mediation as a runtime synthesis problem: given an application utility contract and a CI context, it searches for an executable transformation that preserves task utility while producing a residual information flow accepted by the configured CI rules.

Contextual integrity provides the normative foundation for this decision because it evaluates whether an information flow is appropriate for a context based on actors, attributes, purposes, and transmission principles [8]. However, CI alone does not determine what technical representation should be released. Granting raw access may reveal more information than an application needs, while denying access may prevent useful applications from functioning. We therefore frame privacy mediation as *dynamic preprocessing pipeline synthesis constrained by contextual integrity*. Rather than relying on developers or space operators to manually author a preprocessing pipeline for each application and context, *privMediator* constructs candidate pipelines at runtime from typed preprocessing operators, evaluates the residual information flows they induce, and selects the least-revealing feasible representation ordered by a transparent residual-risk score.

Why this is not just policy compliance. The contribution is not the particular CI rule library used in our experiments. A static policy engine could reject flows that match authored prohibitions, but it would not determine which concrete representation (e.g. raw stream, redacted media, pose, count, event

record, or no release) should be produced for a utility-bearing application. *privMediator* contributes the execution architecture that connects a deployment’s authored rules to typed preprocessing operators, residual-disclosure effects, utility contracts, and least-revealing selection. Another deployment can replace the rule library, residual-risk weights, or operator catalog without changing the synthesis procedure: hospital policies could prioritize clinical-access rules, workplace deployments could add labor or notice constraints, and building operators could add new sensor transformations as versioned operator contracts.

To make runtime synthesis possible, *privMediator* needs machine-readable descriptions of what preprocessing operators do. We introduce a typed operator-effect model in which operators declare the representations they consume and produce, the utility capabilities they preserve, the contextual integrity annotations they contribute, and the residual disclosure effects they induce. These contracts allow the mediator to compose operators at runtime, reason about the information flow produced by a candidate pipeline, and compare feasible representations by residual disclosure. For ambiguous contextual judgments, the system can optionally leverage LLM-based norm approximations as a fallback, though these soft judgments are strictly bounded by *privMediator*’s deterministic CI decision structure and cannot override hard constraints.

This paper makes three contributions. First, it introduces a typed operator-effect model for compositional reasoning about privacy-motivated preprocessing, including output representations, utility capabilities, CI annotations, and residual disclosure effects. Second, it presents a runtime synthesis architecture that generates candidate preprocessing pipelines, checks utility compatibility, evaluates residual CI flows, and selects the least-revealing feasible output or rejects the application request when no feasible release exists. Third, it evaluates *privMediator* across 33 contextual scenarios, four smart-space tasks, fixed-input and flexible-input application contracts, a Prolific [1] appropriateness survey with 82 participants and 1,476 ratings, and ablations of CI filtering and least-revealing selection. Our work shows that contextual integrity can be operationalized as executable representation selection: a mediator can synthesize technical releases, refuse no-feasible contexts, and make utility, disclosure, and refusal tradeoffs auditable without treating the rule library as a universal privacy oracle.

2 Background

Frictional shared smart environments. Figure 1 summarizes the setting we study: sensor-instrumented physical spaces where infrastructure sensors are controlled by a space operator, but the people captured by those sensors may be occupants, visitors, workers, customers, patients, guests, bystanders, or other transient subjects. We call these settings *frictional shared smart environments* because privacy negotiation is conceptually desirable but operationally intractable as a default mechanism. Four properties create this friction: *operator-subject separation*, where the sensing operator differs from the people being sensed;

limited *bystander reachability*, where affected individuals cannot reliably be contacted before or during sensing; weak *preference channels*, where notices, negotiation tools, interfaces, or enforceable opt-outs may be absent or impractical; and *dynamic context*, where appropriateness changes with purpose, time, occupants, activity, or events. These properties motivate mediation mechanisms that decide not only whether a sensor may be used, but what representation of the sensed environment should be released.

Pipelines, utility, and residual disclosure. We frame privacy management in these environments as a problem of selecting a preprocessing pipeline. Rather than granting or denying access to a raw sensor stream, the mediator decides what transformed representation the downstream application should receive. A camera stream might be transformed into a person count, cropped event frame, blurred video, object detections, pose keypoints, or activity label; an audio stream might become a decibel level, sound-event label, or command intent rather than intelligible speech.

We define *utility* as purpose-bound task utility: the extent to which a transformed stream preserves the downstream application’s ability to perform its stated task. This follows the task-performance view common in privacy-preserving data mining, where transformed data is evaluated by whether it still supports useful analysis or accurate models [3, 9, 34]. In our setting, utility is specified by the application request: HVAC may require only room-level occupancy, fall detection may require pose or motion features, and entryway security may require an event notification or minimized verification image.

We use *privacy risk* as an umbrella term for exposure that remains after preprocessing. When precision is needed, we distinguish three related notions. A *residual disclosure vector* is the set of sensitive attributes that remain inferable from a transformed output, such as identity, face, speech content, activity, trajectory, gait, or co-presence. A *residual risk score* is a context-specific scalar metric computed over that vector to rank and compare feasible pipelines by how revealing they are in the current setting. A *residual CI flow* is the contextual-integrity flow induced by a candidate pipeline: the CI actors and transmission principle, together with the residual information types exposed by the transformed output. Thus, the relevant privacy question is not only which sensor was used, but what information remains after preprocessing, who receives it, for what purpose, and under what conditions.

This distinction matters because common transformations reduce some disclosures while preserving others. Face-blurred video may remove facial identity while still revealing clothing, activity, and trajectory; pose keypoints remove pixels and background while potentially revealing gait, disability, exercise, or fall-related behavior; and occupancy counts remove direct identity while potentially revealing routines when collected continuously.

Fixed versus flexible application settings. Many deployed sensing applications have rigid, fixed-input dependencies: a model

Table 1: Terminology used to connect application requirements, operator contracts, residual disclosure, and contextual integrity.

Term	Meaning	Example
Application output type	Technical representation requested or accepted by an application	person_count, pose_keypoints, sound_event_label
Operator utility capability	Task capability preserved or produced by an operator	occupancy estimation, fall detection support, sound-event recognition
Residual disclosure attribute	Sensitive attribute still exposed after preprocessing	face, identity, speech content, activity, trajectory, gait
CI information type	Contextual-integrity description of what information flows to a recipient	occupancy information, coarse presence, activity information, speech information

designed around raw video or audio may not immediately accept a minimized text label, pose sequence, or event record. We call this the *fixed-input setting*. At the same time, some emerging application designs can expose flexible utility contracts, or can be wrapped with adapters that consume semantic representations such as detections, counts, keypoints, sound-event labels, or task events. We call this the *flexible-input setting*. This distinction is an evaluation assumption rather than a claim that all future applications will natively consume arbitrary representations.

We evaluate *privMediator* under both settings. The fixed-input setting reflects deployed applications with brittle raw-like interfaces and tests whether mediation can reduce disclosure without changing the downstream application boundary. The flexible-input setting is forward-looking: it tests the additional benefit available when an application exposes a purpose-bound utility contract or adapter boundary that can accept multiple representations. We do not assume that all current smart-space applications have this property; rather, flexible-input evaluation shows the design target that becomes available when applications are built to negotiate minimized outputs.

Contextual integrity. Utility and residual-risk metrics alone do not determine whether a data flow is acceptable. The same residual disclosure may be appropriate in one context and inappropriate in another: room-level occupancy may be acceptable for office energy management but inappropriate in a bathroom, prayer room, or medical setting. We therefore use contextual integrity (CI) as the normative framework for judging contextual appropriateness. CI models a personal information flow using parameters such as the *data subject*, *sender*, *recipient*, *information type*, and *transmission principle* [41]:

(data subject, sender, recipient,
information type, transmission principle).

To operationalize this framework, our implementation explicitly parametrizes the overarching social context and application purpose alongside these tuples to determine which specific contextual norms apply.

Our CI rules are not treated as a universal normative oracle. Instead, they are a configurable rule library for a particular deployment or experimental setting. Some rules encode hard legal, institutional, or policy constraints; others encode hand-authored contextual norms used to instantiate the scenarios in our evaluation. The configured-rule audit in Section 4 should therefore be read as a policy-compliance and selectivity metric, not as independent ground truth about all possible stakeholder preferences. For ambiguous cases, the system can use bounded norm judgments, and our user study evaluates whether resulting flows align with participant judgments. The key design point is that CI constrains the information flow induced by a technical preprocessing choice: utility determines whether a pipeline can satisfy the application request, the residual disclosure vector describes what remains exposed, the residual risk score ranks feasible outputs by exposure, and the residual CI flow determines whether the resulting disclosure is appropriate for the current context.

3 Method

We design a privacy mediator for smart environments that sits between sensing infrastructure and downstream applications. Rather than granting or denying raw sensor access, the mediator treats each *application request* as a preprocessing-pipeline selection problem. An application request contains a utility contract, which specifies what the application needs to consume, and a CI context, which specifies the contextual-integrity fields used to evaluate the resulting information flow. Given this request, a typed operator-effect catalog, and a CI rule library, the mediator searches for a preprocessing pipeline whose output is useful for the requested task while inducing a residual information flow accepted by the configured rules. If no generated pipeline satisfies these constraints, the mediator returns a *rejected application request*: no transformed representation is released to the application.

The central abstraction is a *typed operator-effect model*. Each preprocessing operator declares the representations it can consume and produce, the task capabilities it supports, the CI terms it contributes, and the residual information that may remain inferable after the transformation. These declarations let the mediator enumerate type-compatible operator chains, check whether outputs match the application contract, propagate residual disclosure estimates, evaluate the induced CI flow, and select a feasible low-disclosure output under the active rule library.

This design avoids two extremes. Coarse access control forces applications into raw allow/deny decisions even when transformed representations such as counts, event labels, keypoints, redacted media, or task decisions would suffice. Unconstrained LLM policy-making lets a model directly decide whether a data flow should be allowed. *privMediator* instead makes the residual information flow explicit: operator contracts define

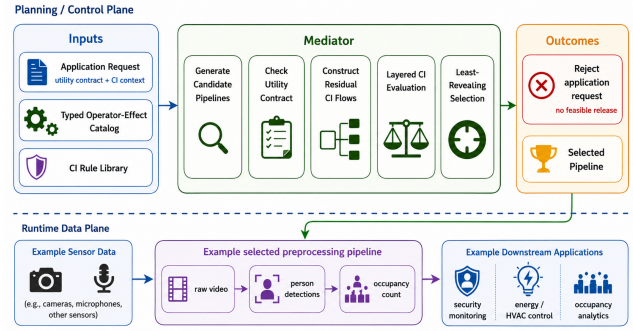


Figure 2: *privMediator* uses structured requests, operator-effect contracts, and CI rules to select a low-disclosure preprocessing pipeline or reject the application request.

the candidate space, CI rules evaluate the induced flow in context, optional LLM judgments are bounded by hard rules, and final selection is performed by a transparent residual-disclosure minimization procedure.

3.1 System Overview

Figure 2 shows the mediator architecture. The implemented mediator operates in a planning/control plane and produces a decision that is applied in the runtime data plane. The planning inputs are an application request, a typed operator-effect catalog, and a CI rule library. Natural-language request normalization and context extraction can be used as an upstream front end, but are outside the evaluated mediator; all experiments assume that requests have already been converted into structured utility and CI fields.

The mediator produces one of two evaluated outcomes: a selected preprocessing pipeline or a rejected application request. A request is rejected when no generated candidate satisfies the active utility, CI, and residual-disclosure constraints. When a selected pipeline exists, it is deployed in the runtime data plane to transform sensor data before release to the downstream application. Candidate generation, utility checking, residual propagation, and final selection are symbolic. Optional LLM use is restricted to bounded norm judgment and cannot enumerate pipelines, override hard rules, or perform final selection. Appendix A.6 summarizes component authority.

Trust and authority boundary. *privMediator* assumes that the mediation layer, operator-effect catalog, CI rule library, and runtime enforcement boundary are trusted and isolated from downstream applications. The evaluated system decides what representation to release once a structured request has reached this trusted boundary; it does not solve malicious sensors, compromised hardware, applications that lie about their purpose, or side channels outside the mediator. This assumption lets the paper focus on the decision procedure for representation release, while Section 6 discusses complementary deployment controls.

3.2 Application Requests and Operator-Effect Contracts

Applications are represented by structured requests rather than raw-stream entitlements. An application request A contains a utility contract U_A and a CI context C_A . The utility contract specifies the requested task capability, acceptable sensing sources, accepted output representations, request-level CI requirements, residual-disclosure bounds, and numerical quality requirements. The CI context provides the sender, subject, recipient, purpose, transmission principles, physical space, and social context used during rule evaluation. For example, a visitor-presence application may accept a detection record, occupancy count, redacted video-like input for a fixed downstream model, or final event label, while also requiring minimization or forbidding identity-bearing outputs. Representative request families are summarized in Appendix A.2.

We model the operator catalog as a typed effect system over representations. Let $\mathcal{T}$ be the set of technical representations, including media representations such as raw video, redacted video, filtered audio, and semantic representations such as detections, pose keypoints, sound-event labels, occupancy records, and task events. Each operator variant $o \in \mathcal{O}$ declares an input type predicate $\tau_{\text{in}}(o)$, an output representation $\tau_{\text{out}}(o)$, utility capabilities $\kappa(o)$, CI annotations $\gamma(o)$, transformation effects $\eta(o)$, and residual-disclosure effects $\delta(o)$. A pipeline candidate $c = (o_1, \dots, o_k)$ is type compatible when each operator consumes the representation produced by the previous operator, subject to the catalog’s cap-matching and schema-adaptor rules. The full vocabulary and additional operator examples are summarized in Appendix A.3.

A candidate is symbolically utility-compatible when its final output representation matches one of the accepted output caps in U_A and its accumulated utility capabilities support the requested task capability. Numerical quality requirements, such as latency, recall, precision, or count error, are not proved by symbolic planning; they are evaluated separately in the downstream utility experiments.

The residual disclosure state is a vector over sensitive attributes. Let $\mathcal{D}$ be the evaluated set of tracked residual attributes (e.g., *identity*, *face*, *gait*, *speech content*, *trajectory*, and *aggregate presence*). Each candidate has a residual vector $\mathbf{d}_c \in L^{|\mathcal{D}|}$, where $L = \{\text{none}, \text{low}, \text{medium}, \text{high}, \text{unknown}\}$ is ordered as $\text{none} < \text{low} < \text{medium} < \text{high} < \text{unknown}$. The implementation maps these levels to ordinal values $\rho(\text{none}) = 0$, $\rho(\text{low}) = 1$, $\rho(\text{medium}) = 2$, $\rho(\text{high}) = 3$, and $\rho(\text{unknown}) = 4$.

Operators compose by abstract interpretation over this residual vector. For an operator o , the next residual vector is computed from the current vector $\mathbf{d}$ and the operator’s residual effect $\delta(o)$. Media-preserving operators use a *preserve* default: unmentioned attributes remain unchanged. Semantic summarization operators may use a *drop* default, resetting unmentioned raw-media attributes to none before explicit effects are applied. Join and fusion operators combine streams conservatively using attribute-wise maximum, and may add correlation risks such as co-presence or trajectory.

Table 2: Example operator-effect contract. The mediator uses the contract to reason about both utility and residual disclosure before execution.

Contract field	Example value
Input representation	raw video frames or video windows
Output representation	pose-keypoint sequence
Utility capabilities	fall-detection and activity-recognition support
Transformations	removes raw pixels, face texture, and background imagery
Residual disclosure vector	face: none; identity: low/medium; body shape: medium; gait: medium; activity: high; location: medium; trajectory: medium
CI annotations	data minimization, semantic feature release, no raw-video release

The implementation applies explicit effects in a fixed order. set assigns a declared risk level; remove sets an attribute to none; reduce lowers a risk level according to the declared reduction rule; and add conservatively raises an attribute when fusion or correlation may reveal additional information. Thus, residual propagation is deterministic composition over typed operator effects, not an LLM judgment.

Table 2 illustrates one contract; the residual vector records what remains inferable after preprocessing rather than treating a transformation as categorically private. These contracts are authored, versioned artifacts rather than learned privacy guarantees. The implementation therefore treats unknown or unvalidated effects conservatively, and the evaluation interprets residual-disclosure estimates as auditable engineering assumptions rather than formal leakage bounds.

3.3 Candidate Generation and Residual CI Flows

Candidate generation is a bounded symbolic search, not an LLM planning step. The operator catalog induces a directed multigraph $G = (V, E)$ in which vertices V are technical representations and edges E are operator variants that transform one representation into another. The planner searches over type-and-effect states rather than representations alone, because two paths may produce the same output representation while inducing different residual disclosures, CI terms, transformations, or utility capabilities.

A planner state is a tuple

$$s = (\tau, \mathbf{d}, K, \Gamma, P),$$

where τ is the current representation, $\mathbf{d}$ is the current residual disclosure vector, K is the accumulated set of utility capabilities, Γ is the accumulated set of CI annotations and transmission principles, and P is the operator sequence. The planner initializes source states from source variants that satisfy A ’s source requirements, then performs a goal-directed breadth-first search over operator variants whose input caps match the current representation. The search is bounded by a maximum depth and maximum number of expanded states.

A state emits a candidate when its current representation matches an accepted output cap in U_A and its accumulated capabilities satisfy the requested task capability. The planner may apply request-level prefilters, such as accepted-cap transformation constraints or residual-disclosure bounds, but these

prefilters are not the final authorization decision; the layered CI evaluator and final selector still evaluate emitted candidates.

For each emitted candidate c , the mediator constructs a residual CI flow

$$f_c = (C_A, \tau_c, \mathbf{d}_c, \Gamma_c, \sigma_c),$$

where C_A supplies the sender, subject, recipient, purpose, context, space, and transmission principle; τ_c is the final output representation; $\mathbf{d}_c$ is the candidate’s residual disclosure vector; Γ_c contains accumulated CI annotations; and σ_c records the pipeline stage. Residual attributes are mapped into CI-relevant information types: speech-content residuals become speech-related information types, aggregate-presence residuals become occupancy information, and trajectory residuals become inferred movement information. Appendix A.3 summarizes the staged-flow vocabulary.

The implementation preserves staged flows by default. We use four stage labels—raw input, intermediate representation, retained artifact, and output to application—so hard rules can distinguish transient in-hub processing from artifacts that are retained or released. The no-staged-flows ablation collapses these labels to the output boundary.

3.4 Layered CI Evaluation and Probe Validation

The CI evaluator applies rule-library constraints to each residual flow f_c in layers. Classification rules first add tags to the residual flow. Hard denial rules reject prohibited flows. Required-condition, required-transformation, and conditional-allowance rules check whether the flow contains necessary transmission principles, contextual fields, transformations, or forbidden conditions. Preference-only rules may inform diagnostics or ranking, but are not treated as hard accept/reject decisions. Representative CI rules are shown in Appendix A.4.

Rule authoring and maintenance. The rule library is a deployment artifact. In our prototype it is hand-authored from contextual integrity scenario requirements and public policy or guidance examples, and each rule records provenance metadata: a source name, source type, natural-language policy statement, and, for external sources, a URL. The evaluated libraries draw from short-term-rental platform policies (Airbnb and Vrbo), campus and workplace surveillance policies, and legal, regulatory, or professional guidance for schools, research, and clinical settings; a small number of created templates capture mediator-specific boundaries such as final-task-decision outputs. We use these sources as representative policy constraints for evaluating the mediation architecture, not as a complete legal or platform-compliance implementation.

The rules fall into four operational families: *hard denials*, such as hidden recording devices or cameras in prohibited private spaces; *required conditions*, such as disclosure, explicit consent, authorized-personnel access, or limited retention; *required transformations*, such as face redaction, speech-content removal, or aggregation before release; and *preference-only rules*, which may affect diagnostics or ranking but do not authorize

or deny by themselves. In a deployment, these rules would be authored and revised by the organization or governance body responsible for the space, potentially with input from legal, compliance, resident, worker, patient, or research-ethics stakeholders. Rules need not change for every application: they change when policy, social context, sensor placement, or acceptable transmission principles change, while new technical capabilities are added through operator contracts.

If a candidate fails a hard rule, it is rejected before any LLM judgment is considered. For candidates that pass hard rules, the mediator may invoke an LLM as a bounded norm interpreter over the structured residual flow. In normal operation, the LLM cannot override hard rules, enumerate pipelines, or perform final selection; an LLM-only mode exists only as an ablation. The CI rule library is configurable rather than a universal normative oracle, so deployments can replace or extend the rules with institutional policy, legal constraints, or survey-derived norms without changing the synthesis algorithm.

The mediator also maintains a metadata residual estimate $\mathbf{d}_c^{\text{meta}}$ from operator contracts. When transformed artifacts and a probe configuration are available, empirical privacy probes estimate inferability from the concrete transformed output. These probes are implemented as auxiliary detectors or recognizers over transformed artifacts, such as face detection, person/body detection, OCR, scene-object probing, audio-presence checks, ASR-based speech-content checks, diarization, and aggregate or trajectory inference probes. They do not directly authorize a flow; they only produce a probe residual vector $\mathbf{d}_c^{\text{probe}}$.

Probe evidence is combined conservatively with metadata:

$$d_{c,i} = \max(d_{c,i}^{\text{meta}}, d_{c,i}^{\text{probe}}) \quad \forall i \in \mathcal{D},$$

where the maximum is taken under the residual-risk ordering above. Thus probe evidence can make a candidate appear more revealing, but cannot make it appear safer than its symbolic contract. The probes are implementation checks for concrete artifacts, not formal privacy proofs against arbitrary observers. When no probe artifacts are supplied, final selection uses $\mathbf{d}_c^{\text{meta}}$ alone; Appendix A.8 gives additional probe-validation details.

3.5 Least-Revealing Feasible Selection and Rejection

The final selector consumes generated candidates, CI evaluations, optional probe results, and the application request. Let $\mathcal{C}$ be the set of generated candidates. The selector first forms the feasible set:

$$\mathcal{C}_{\text{feas}} = \{c \in \mathcal{C} : U(c, A) = 1 \wedge \text{CI}(f_c, N) = \text{accept} \wedge \mathbf{d}_c \preceq \mathbf{b}_A\},$$

where $\mathbf{b}_A$ is the request’s residual-disclosure bound and $\preceq$ is attribute-wise comparison under the residual-risk ordering.

Among feasible candidates, *privMediator* selects the lowest-disclosure feasible candidate according to a context-specific residual-risk score. Let $\mathbf{w}_{CA}$ be the policy weight vector active for the request context, where $w_{CA,i}$ encodes how strongly the

deployment penalizes residual attribute i in this purpose and context. The residual-risk score is

$$R_{CA}(c) = \sum_{i \in \mathcal{D}} w_{CA,i} \cdot \rho(d_{c,i}).$$

The default selector chooses

$$c^* = \arg \min_{c \in C_{\text{feas}}} R_{CA}(c).$$

Deterministic tie breakers (including lexicographic attribute ordering $\text{Lex}(\mathbf{d}_c)$, pipeline latency, and application output priority) are applied sequentially if multiple candidates share the minimal risk score. The residual-risk weights are transparent policy parameters, not learned constants or universal privacy claims; in our experiments, they are hand-authored and tested through the ablations in Section 4.5.

If C_{feas} is empty, *privMediator* returns a *rejected application request*: no transformed representation is released. Diagnostics distinguish no utility-compatible representation, hard denials, residual-disclosure failures, and unresolved constraints. Deployments could route some rejected or uncertain requests to consent, human review, or degraded-task negotiation, but our evaluation treats them as rejected application requests.

3.6 Algorithm

Algorithm 1 summarizes the implemented procedure.

4 Evaluation

We evaluate whether *privMediator* can mediate smart-space sensing requests by preserving downstream utility while enforcing configured CI constraints and reducing residual disclosure under operator contracts. The evaluation has five purposes: decision coverage, downstream utility, human judgments of perceived contextual appropriateness, ablations, and deployment-oriented context-switch latency. Throughout this section, CI-conflict and hard-rule-violation counts are *configured-rule audit metrics*: they measure whether a method releases flows that violate the authored rule library used in the experiment. They should be interpreted as policy-compliance and selectivity metrics, not as independent ground truth about all possible stakeholder preferences.

RQ1: Synthesis and feasibility. Does the mediator select a utility-compatible output when one is contextually acceptable, and reject the application request when no feasible release exists?

RQ2: Downstream utility. Conditional on selecting an output, do generated preprocessing pipelines preserve task performance relative to raw input and manually tuned preprocessing?

RQ3: Perceived contextual appropriateness. Are less revealing mediated information flows judged more appropriate than raw or utility-maximizing outputs, and how do method-level ratings change once rejected contexts are considered?

RQ4: Design choices. Which mediator components affect selection rate, utility retention, and contextual appropriateness?

Algorithm 1 CI-constrained preprocessing pipeline mediation

Require: application request A , operator catalog O , CI rules N , optional probe artifacts T

Ensure: selected preprocessing pipeline or rejected application request

- 1: Load structured utility contract U_A and CI context C_A from A
- 2: Construct source states and operator variants from typed operator-effect contracts in O
- 3: Run bounded BFS over type-and-effect states $s = (\tau, \mathbf{d}, K, \Gamma, P)$
- 4: Emit candidates whose output caps and accumulated capabilities satisfy U_A
- 5: **for** each candidate c **do**
- 6: Construct residual CI flow $f_c = (C_A, \tau_c, \mathbf{d}_c^{\text{meta}}, \Gamma_c, \sigma_c)$
- 7: Evaluate f_c using CI rules N and optional bounded norm judgment
- 8: **if** probe artifacts T are available for c **then**
- 9: Run empirical probes to estimate $\mathbf{d}_c^{\text{probe}}$
- 10: Set $\mathbf{d}_c \leftarrow \max(\mathbf{d}_c^{\text{meta}}, \mathbf{d}_c^{\text{probe}})$ attribute-wise
- 11: **else**
- 12: Set $\mathbf{d}_c \leftarrow \mathbf{d}_c^{\text{meta}}$
- 13: **end if**
- 14: **end for**
- 15: Build feasible set $C_{\text{feas}} = \{c : U(c, A) = 1 \wedge \text{CI}(f_c, N) = \text{accept} \wedge \mathbf{d}_c \preceq \mathbf{b}_A\}$
- 16: **if** $C_{\text{feas}} = \emptyset$ **then**
- 17: **return** rejected application request with diagnostics
- 18: **end if**
- 19: Select $c^* = \arg \min_{c \in C_{\text{feas}}} R_{CA}(c)$ with deterministic tie breakers
- 20: **return** selected preprocessing pipeline c^*

RQ5: Dynamic context switching. Can the mediator recompute decisions quickly enough for context changes, and does it avoid hard-rule-violating releases during those changes?

4.1 Experimental Setup

Tasks and datasets. We evaluate utility on four smart-space tasks that match the task categories in our contextual-integrity scenario pool: visitor presence detection, fall detection, activity-of-daily-living (ADL) recognition, and domestic sound monitoring. Table 3 gives the dataset, downstream interface, and primary metric for each task; Appendix B.2 gives dataset-specific preprocessing and evaluator details.

Context scenarios. We instantiate 33 contextual-integrity scenarios across the four tasks: eight visitor-presence scenarios, eight fall-detection scenarios, eight ADL-recognition scenarios, and nine domestic-sound scenarios. Each scenario fixes the context, space, sender, subject, recipient, purpose, and transmission principle, but leaves the released representation unspecified. Scenario identifiers are not contiguous because pilot and ambiguous child/guardian scenarios were removed

Table 3: Downstream utility tasks, datasets, interfaces, and primary metrics.

Task	Dataset	Downstream model/input	Primary utility metric
Visitor presence detection	ChokePoint [50]	Person detector over image/video frames; predictions aligned to frame-level person annotations	Frame-level person-presence F1
Fall detection	Le2i Fall [12]	Pose-based fall classifier over keypoint sequences/windows	Video-level macro-F1
ADL recognition	YouHome ADL [42]	Audio-visual ADL classifier over synchronized image frames and audio waveform segments	Macro-F1 over ADL classes
Domestic sound monitoring	CHiME-Home [23]	Multi-label audio tagger over fixed-duration audio chunks	Macro-F1 over sound labels

after survey-design feedback; for example, the final pool contains 33 scenarios even though one retained identifier is S034.

Application contracts. We evaluate two application-interface assumptions. In the *fixed-input* setting, the downstream application expects the representation used by the existing task evaluator, such as image/video frames, pose-compatible inputs, synchronized audio-video, or audio waveforms. In the *flexible-input* setting, the application declares several acceptable output representations, including semantic outputs such as detections, occupancy counts, pose keypoints, sound-event labels, or event records. Appendix A.2 and Appendix B.2 describe the request schemas and adapter boundaries (Tables 9 and 23).

Compared methods. We compare RAW, a developer-written MANUAL preprocessing policy, a DIRECT-LLM baseline, and *privMediator*. RAW forwards the original sensor stream. MANUAL applies a fixed task-specific preprocessing policy without context-specific CI reasoning. DIRECT-LLM selects or refuses a pipeline using a prompt over the application request and context. *privMediator* performs symbolic candidate generation, utility checking, residual-flow construction, CI filtering, and least-revealing feasible selection. We also evaluate ablations that remove CI filtering, optimize utility only, or alter the final selection objective. The fixed and flexible operator catalogs contain 25 and 28 operators, respectively; selected pipelines use 23 distinct operator IDs, all present in the corresponding catalogs.

Hardware. Experiments were run on a workstation with an NVIDIA GeForce RTX 2070 GPU and an AMD Ryzen 7 3800X 8-core processor with 16 hardware threads. GPT-4o-mini [26] was used as the LLM for both the direct LLM baseline as well as *privMediator*.

4.2 Pipeline Synthesis and Feasibility

This experiment asks whether each method selects an output and whether it releases data in contexts where the configured rule library says no feasible release should exist. We use *Rejected* for cases where the method returns no output because no generated pipeline satisfies the active utility, CI,

Table 4: Contract-derived residual disclosure for selected outputs. Values are descriptive metadata audits, not formal leakage guarantees.

Run	Method	Sel.	Raw-like	Mean risk	Face $\geq$ med. / Speech $\geq$ med.
Fixed	RAW	33	25	27.5	72.7% / 51.5%
Fixed	MANUAL	33	0	18.0	0.0% / 15.2%
Fixed	<i>privMediator</i>	18	0	17.4	0.0% / 0.0%
Flexible	RAW	33	25	26.3	72.7% / 51.5%
Flexible	MANUAL	33	0	16.3	0.0% / 15.2%
Flexible	<i>privMediator</i>	21	7	14.4	19.0% / 14.3%

and residual-disclosure constraints. We use *Review* only for the direct-LLM baseline when it explicitly asks for human review rather than selecting or rejecting.

Figure 3a summarizes decisions across the 33 context scenarios. RAW and MANUAL always select an output. DIRECT-LLM selects an output in 28 scenarios and returns no output in five cases: four rejected requests and one review decision. *privMediator* selects an output in 18 scenarios and rejects 15 requests. These rejections are mediation decisions, not execution failures: they indicate that, under the configured rule library and residual bounds, the mediator found no utility-compatible representation that also satisfied the active CI and residual-disclosure constraints. RAW releases raw video, audio, and AV streams; MANUAL, DIRECT-LLM, and *privMediator* primarily release redacted images, pose, filtered audio, or redacted/filter AV outputs when they select a pipeline.

Residual-disclosure accounting. We also audit residual disclosure and rejection causes. Contract-derived accounting shows that transformed outputs reduce raw-like exposure, face exposure, and speech-content exposure relative to RAW. Table 4 gives the compact main-body summary: in the fixed-input run, *privMediator* releases no raw-like outputs, no selected outputs with medium-or-higher face exposure, and no selected outputs with medium-or-higher speech-content exposure. These values are metadata audits over typed operator effects, not formal leakage guarantees; empirical probes are used as conservative implementation checks that can raise, but not lower, residual-risk estimates. Appendix B.1 reports the full residual-disclosure vectors and probe examples.

The refusals also reflect substantive contextual constraints rather than missing implementations. None of the 15 *privMediator* rejections was dominated by lack of utility support: each had at least one utility-compatible representation, but all feasible representations failed a hard denial, required condition, or residual-disclosure constraint. Four refusals involved hidden or undisclosed monitoring, four involved private or sensitive spaces, and the remaining seven reflected routine surveillance, research/bystander limits, outdoor-location conditions, or indoor-audio constraints. This matters because a rejection is not equivalent to a failed pipeline run; it is the mechanism by which the mediator surfaces a conflict between application utility and configured contextual constraints. Appendix B.1 gives the full rejection diagnostics and representative generated records.

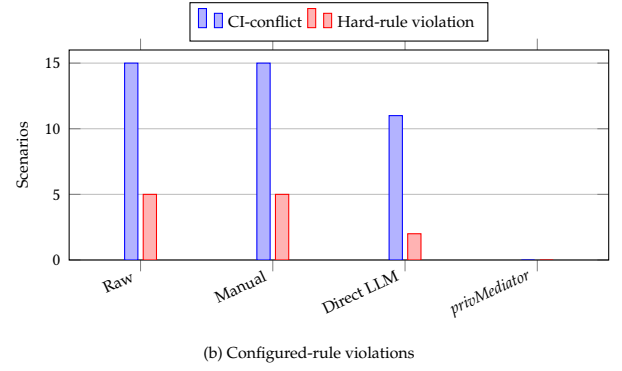
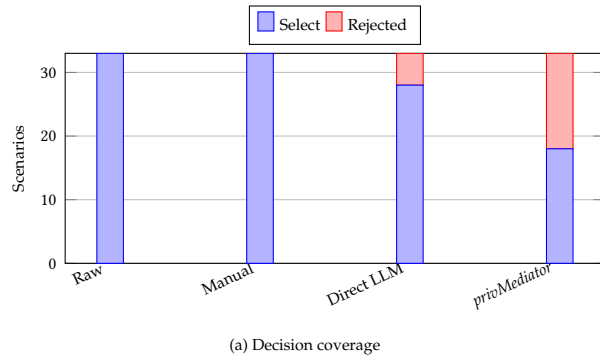


Figure 3: Coverage and configured-rule audit outcomes across 33 scenarios. Direct-LLM review is counted with rejected decisions because no data is released. CI-conflict and hard-rule-violation releases are fixed-input outputs that violate the configured CI audit rules.

Table 5: *privMediator* selected-output utility by task. Utility is conditional on release and normalized to RAW.

Task	Fixed sel.	Fixed ret.	Flex sel.	Flex ret.
Visitor	4/8	0.973	7/8	0.858
Fall	6/8	1.000	6/8	0.804
ADL	4/8	1.021	4/8	1.021
Sound	4/9	0.866	4/9	0.966

4.3 Downstream Utility

We report utility conditional on a method selecting an output, and report selection/rejection counts separately. Utility retention is the primary task metric divided by the corresponding RAW metric; it is not coverage-adjusted, so it measures whether released outputs remain task-useful while leaving the availability cost visible. Fixed-input outputs are evaluated with the existing task evaluators, while flexible semantic outputs use the adapter boundary declared in the application contract; Appendix B.2 gives the evaluator and adapter details.

Across four tasks, *privMediator* retains 96.5% fixed-input selected-output utility and 91.2% flexible-input selected-output utility. Table 5 shows that the utility tradeoff is not uniform across tasks. In the fixed-input setting, selected outputs retain manual-policy utility whenever a CI-feasible representation exists. In the flexible-input setting, Visitor and Fall lose selected-output utility because the mediator chooses more minimized outputs such as counts, detections, or pose key-points; ADL remains stable, and Sound retains 96.6% of raw utility when selected. Appendix B.1 reports the full per-task fixed and flexible utility tables.

Table 6 gives the combined coverage-aware view. It shows the tradeoff that selected-output utility alone obscures: always-releasing methods retain high availability but release in configured CI-conflict contexts, while *privMediator* reduces availability by refusing no-feasible-output scenarios and retains high utility on selected outputs.

Figure 3b summarizes the coverage-aware interpretation for the fixed-input run across the full context pool. Selected-output utility alone can be misleading: methods that always release data can obtain high utility by releasing in contexts where release should not occur. RAW and MANUAL each release in 15 CI-conflict contexts, including 5 hard-rule-violating releases. DIRECT-LLM reduces but does not eliminate these releases, selecting outputs in 11 CI-conflict contexts, including 2 hard-rule-violating releases. *privMediator* has lower selection rate, but eliminates CI-conflict and hard-rule-violating releases.

Selected-output utility shows that *privMediator*’s released pipelines remain task-useful; selection rate and configured-conflict counts show the complementary cost and benefit of refusing outputs in contexts where the rule library finds no CI-feasible release.

4.4 Perceived Contextual Appropriateness Survey

We evaluate perceived contextual appropriateness with a Prolific [1] survey in which participants rated concrete scenario-output information flows from the perspective of the person the data is about. Items combined a CI context with a shared output; method labels were hidden. The survey measures perceived appropriateness, not objective leakage, legal compliance, or consent validity. The completed-session analysis includes 82 participants and 1,476 analyzed ratings over deduplicated scenario-output pairs; Appendix B.3 gives the interface, filtering, randomization, output-class summaries, and full method-linked table (Table 21).

privMediator receives the highest descriptive method-linked mean rating among the principal methods (3.72, with 66.0% of ratings at least 4), but differences among transformed outputs are smaller than the difference between raw and transformed outputs. A mixed-effects model finds that raw outputs are rated lower than the transformed-output reference condition (coefficient $-0.64, p < 0.001$), while differences among *privMediator*, DIRECT-LLM, and MANUAL selected outputs

Table 6: Coverage-aware summary for principal methods. Utility is selected-output retention averaged over task-level primary metrics; configured-rule conflicts are fixed-input audit counts.

Method	Fixed sel.	Flex sel.	Fixed util.	Flex util.	CI-conflict rel.	Hard-rule viol.	Survey mean
RAW	33/33	33/33	1.000	1.000	15	5	2.83
MANUAL	33/33	33/33	0.965	0.927	15	5	3.43
DIRECT-LLM	28/33	33/33	0.965	0.822	11	2	3.56
<i>privMediator</i>	18/33	21/33	0.965	0.912	0	0	3.72

Table 7: Key ablations. CI filtering controls refusal; least-revealing selection controls which feasible output is released.

Method	Fixed sel.	Fixed util.	Survey mean	Approp.
<i>privMediator</i>	18/33	0.965	3.72	66.0%
UTILITY-ONLY	33/33	1.000	2.97	44.2%
NO CI FILTER	33/33	0.965	3.45	59.5%
NO LEAST-REVEALING	18/33	0.989	3.31	52.5%
LATENCY-FIRST	18/33	0.989	3.31	52.5%

are comparatively small. We therefore interpret the survey as supporting the contextual plausibility of transformed releases, while the configured-rule audit captures *privMediator*’s distinct policy-level selectivity: DIRECT-LLM still releases in 11 fixed-input CI-conflict contexts, including 2 hard-rule-violating releases, whereas *privMediator* eliminates those releases.

4.5 Ablations and Dynamic Context Switching

The ablations isolate two effects. First, CI filtering controls refusal: UTILITY-ONLY and NO CI FILTER select outputs in every context and retain high selected-output utility, but do so by releasing data in contexts where the full mediator rejects the request. Second, least-revealing selection controls what is released after feasibility filtering: NO LEAST-REVEALING and LATENCY-FIRST keep the same refusal decisions as *privMediator* but choose more revealing feasible outputs, increasing utility retention while lowering appropriateness ratings. Table 7 gives the main-body comparison, and Appendix B.1 reports the full ablation table.

To check whether mediation is practical under changing contexts, we replayed 17 context phases across four dynamic traces. *privMediator* recomputed decisions when the CI tuple changed rather than per frame, selected outputs in 12 phases, rejected or reviewed 5, and made zero hard-rule-violating releases. RAW and MANUAL released in all phases but produced hard-rule-violating releases in 4/17 phases; DIRECT-LLM produced hard-rule-violating releases in 2/17. Median *privMediator* context-switch latency was 2.28s, dominated by candidate generation rather than CI rule checking, and the full dynamic breakdown is in Appendix B.4 (Tables 28, 29, and 30).

5 Related Work

Privacy tensions and contextual norms in smart environments. Prior work shows that smart-environment privacy is a social and contextual problem involving device owners, household

members, guests, workers, and bystanders. Studies of smart-home users and incidental users identify concerns about sensing infrastructure owned or configured by others [4, 15, 51], while work on guests and bystanders shows that expectations depend on relationships, familiarity with the space, and the social cost of raising privacy concerns [39, 46]. Contextual integrity has been used to measure smart-home privacy norms and interpersonal data-sharing expectations [7, 25], and to encode legal rules or privacy preferences for automated reasoning over information flows [2, 7, 8]. This work motivates our use of CI, but it primarily tells us which information flows may be appropriate. Our contribution is complementary: we operationalize CI as a runtime synthesis problem, deciding which executable preprocessing pipeline can produce a utility-preserving residual flow that satisfies those contextual constraints.

Privacy awareness, notice, and preference learning. Another line of work seeks to make privacy practices more visible or easier to act on. Privacy labels communicate IoT or app data practices to users, but prior work shows that labels can be difficult to design, incomplete, or dependent on presentation format [10, 19, 35]. Privacy assistants and notice-and-choice tools provide more interactive support by learning preferences, delivering notices, or helping users make privacy decisions in mobile and IoT settings [16, 17, 21, 33, 38]. These systems improve awareness and preference expression, but they generally assume that privacy decisions can be made by or for reachable individuals. Frictional shared smart environments weaken this assumption: affected people may be transient, unknown, socially unable to object, or sensed together with others. Our system therefore focuses on reducing unnecessary disclosure through mediation when direct notice or negotiation is unavailable or insufficient.

Managing privacy and control in shared environments. Several systems address multi-user control in smart environments. Prior work studies how households coordinate and contest control over smart-home devices and proposes mechanisms for access control or conflict resolution among multiple users and devices [24, 45]. Other systems explore collective or negotiated control: Hivemind supports democratic control of public actuators [32], TEO provides ephemeral ownership so stakeholders can control data generated while they are present [52], and ThingPoll supports negotiation of privacy settings for shared sensing devices [53]. These systems recognize that privacy decisions in shared environments cannot be made solely by device owners. Our work targets a different point in the

design space: cases where affected stakeholders cannot reliably be reached or coordinated. Rather than depending on negotiation as the primary mechanism, the mediator uses contextual constraints and operator contracts to select a least-revealing feasible representation, or reject the application request when no acceptable release exists.

Privacy-preserving sensing and data-processing middleware. A complementary line of systems protects privacy by mediating how applications access, transform, and transmit sensor data. DARKLY interposes on computer-vision inputs and applies privacy-preserving transformations before perceptual applications receive visual data [27], while ipShield enforces context-aware privacy policies by mediating access to sensed information [11]. FlowFence, SmartAuth, and ContextIoT constrain IoT application behavior through declared data flows, semantic authorization, or context-aware prompts [22, 28, 47]. Other systems use hubs, gateways, or middleware to filter, transform, or minimize data before it leaves the local environment, including Privacy Mediators, TIPPERS, PFirewall, Peekaboo, and PrivGuard [14, 18, 29, 40, 49]. These systems show that privacy can be enforced in the data path rather than only through notice or policy text. However, they typically enforce declared flows, apply fixed policies or fixed transformations, mediate known data types, or minimize outgoing data under a predetermined policy. *privMediator* instead synthesizes preprocessing pipelines at runtime from typed operator-effect contracts, checks whether candidate outputs satisfy an application utility contract, evaluates the residual CI flow, and selects the least-revealing feasible representation.

LLM-assisted privacy reasoning. Recent work uses large language models to parse privacy policies, reason about legal requirements, and assist with privacy-preserving data processing [13, 31, 37, 43, 44]. Other work connects LLMs to contextual integrity or privacy-preserving function selection, using models to summarize information, judge whether disclosures are appropriate, or select privacy-preserving transformations from natural-language requirements [30, 36, 48]. These works suggest that LLMs can help bridge informal privacy expectations and executable mechanisms. Our system uses LLMs only in bounded roles, such as optional upstream request normalization or structured norm judgment. LLMs do not enumerate pipelines, override hard CI rules, or perform final authorization. The enforceable decision structure remains symbolic: typed contracts define the candidate space, CI rules constrain the residual information flow, and the final selector chooses among feasible candidates or rejects the request.

6 Discussion

Trusted mediation boundary. *privMediator* assumes a trusted mediation layer between infrastructure sensors and downstream applications. The mediator, operator-effect catalog, CI

rule library, and runtime enforcement boundary must be isolated and trusted. While the system reduces unnecessary exposure by transforming data before release, it does not prevent all adversarial behavior; a malicious application might combine permitted outputs with out-of-band side channels, misstate its purpose before the request reaches the mediator, or infer sensitive facts from repeated permitted outputs. A compromised sensor could also bypass the planning plane entirely. Practical deployments therefore require complementary infrastructure security, including sensor attestation, application authentication, audit logging, access revocation, and sandboxed execution. Our contribution is not a complete trusted-computing architecture, but rather the auditable decision procedure for what data representation should be released when such a mediation boundary is enforced.

Beyond policy compliance. A configured-rule audit by itself would only show that a method follows a policy. *privMediator* is intended to address the harder systems question of how an authored policy becomes an executable, utility-aware release decision. The same rule library can admit multiple technically valid outputs with different utility and disclosure profiles; conversely, a rule violation may remain even after strong redaction if the context forbids the underlying sensing or recipient relationship. The contribution is therefore the coupling of rule evaluation with typed operator synthesis, residual-disclosure propagation, and least-revealing feasible selection, not the claim that our experimental rule library is the only correct set of norms.

Normative scope of configured CI rules. The CI rule library, residual-risk weights, and operator contracts are authored artifacts rather than universal definitions of privacy. In practice, these artifacts can be derived from institutional regulations, workplace policies, building rules, community standards, or survey-derived norms. This separation intentionally decouples the synthesis procedure from the source of normative authority. However, it also means that configured-rule violations in our evaluation are best understood as policy-compliance and selectivity metrics: they show whether a method releases flows that violate the authored rules used for the experiment, not whether all stakeholders would necessarily agree with those rules. We therefore pair the configured-rule audit with selected-output utility, residual-disclosure accounting, probe sanity checks, ablations, and participant appropriateness ratings. The authoring burden is primarily a deployment/governance task rather than an application-developer task: rules should be revised when policies, spaces, sensor placements, or transmission principles change, while ordinary applications interact with the mediator through utility contracts. In our evaluation, rejection diagnostics were explained by recurring rule families—hidden recording, private-space bans, required notice/consent, limited access/retention, and outdoor/security-purpose conditions—rather than by per-application handcrafted decisions.

Specification and contract verification. System efficacy depends heavily on the accuracy of operator contracts and residual-disclosure estimates. Misconfigured operator profiles could understate leakage, while overly conservative profiles could cause unnecessary application rejections. Operator contracts should therefore be versioned, testable, and auditable. The residual-disclosure accounting in Section 4 is useful as a sanity check because it summarizes the exposure implied by these contracts across selected outputs, but it should not be read as a formal privacy or leakage guarantee. Our empirical privacy probes mitigate this risk by detecting unmodeled leakage in concrete transformed artifacts, but they function as implementation checks rather than formal privacy proofs; probe evidence can conservatively raise a candidate’s risk profile, but cannot guarantee bounds against an unmodeled, stronger adversarial observer. A stronger deployment would include independent red-team testing, residual-disclosure regression tests, and review procedures for adding new operators or changing risk weights.

Operational costs of refusal. By making infeasibility explicit, *privMediator* avoids silent releases of data that violate the configured policy. Strict refusal also introduces operational costs in safety-critical settings such as fall detection, clinical monitoring, or security response. Real-world deployments must decide how to handle these explicit rejections, for example by routing them to human-in-the-loop review, emergency escalation, conditional consent, or degraded fallback tasks. *privMediator*’s role is to surface these conflicts and emit auditable diagnostics; exception handling remains a deployment-specific governance decision.

Flexible application contracts. The flexible-input evaluation demonstrates the potential benefit of applications that can accept multiple minimized representations, but it should not be read as a claim that all smart-space applications are already flexible. Some applications will remain bound to raw-like interfaces or will require new adapters before they can consume semantic outputs such as counts, keypoints, or event records. The flexible setting is therefore a design target: it shows what privacy mediation can gain when downstream applications expose purpose-bound utility contracts instead of hard-coded raw-stream dependencies.

Limitations and generalizability. Our user study evaluates perceived contextual appropriateness rather than legal compliance, consent validity, or quantitative information leakage. Our utility evaluation is bounded by four smart-space tasks, 33 scenarios, and a finite operator catalog. Extending *privMediator* to other domains requires expanding the operator schemas, adapters, and CI rule libraries, and validating the resulting residual-disclosure estimates. Nevertheless, the central architecture—synthesizing typed preprocessing pipelines, evaluating residual CI flows, and selecting the least-revealing feasible representation—is not specific to the evaluated datasets and can be instantiated in other sensor-saturated shared environments.

7 Conclusion

This paper presented *privMediator*, a privacy mediator that operationalizes configured contextual-integrity constraints as runtime synthesis over typed preprocessing operators. Rather than treating smart-space privacy as a binary allow/deny decision, *privMediator* dynamically generates candidate preprocessing pipelines, checks them against an application’s utility contract, evaluates the residual CI flows they induce, and selects the least-revealing feasible representation using a transparent residual-risk metric. Across 33 scenarios and four smart-space tasks, *privMediator* preserved selected-output utility when feasible while making zero configured-rule-violating releases, explicitly refusing requests where the rule library found no acceptable representation. Our user study further shows that *privMediator*’s selected representations received higher average appropriateness ratings than raw streams or MANUAL preprocessing policies, while the configured-rule audit captures the complementary benefit of refusing release in no-feasible-output contexts. The value of *privMediator* is therefore its context-sensitive selectivity: it provides an auditable runtime architecture for making utility, disclosure, and refusal tradeoffs explicit in shared smart environments, rather than claiming a universal privacy guarantee.

Ethical Considerations

Our user study asked participants to evaluate the appropriateness of information sharing in hypothetical smart-space scenarios. The study was reviewed by the authors’ institutional review board and certified as exempt from IRB review under 45 CFR § 46.104(d)(2) (Exempt Category 2). Participants provided informed consent before beginning the survey, and responses were analyzed in aggregate.

Open Science

Upon acceptance, we plan to release the *privMediator* codebase, including the mediator implementation, operator-effect catalog, CI rule examples, baseline and ablation configurations, and scripts for reproducing the pipeline-generation and utility-evaluation results. We will also release evaluation meta-data such as scenario definitions, application request templates, generated pipeline summaries, and dataset split files. We will not redistribute third-party datasets; instead, we will provide instructions for obtaining them from their original sources and applying our splits.

For the survey, we plan to release the survey instrument, participant-facing scenario text, scenario-output mappings, and analysis scripts. We will release de-identified participant response data only if permitted by our IRB determination, consent materials, and institutional data-sharing requirements; otherwise, we will release aggregate statistics and derived analysis tables.

AI Use

AI-assisted tools were used during the preparation of this manuscript and research artifacts. ChatGPT was used to improve the clarity and presentation of the writing, including revising awkward phrasing, suggesting organizational improvements, and providing general feedback. Figure 1 was generated with assistance from ChatGPT and subsequently edited manually in PowerPoint. Figure 2 was manually authored and lightly revised with ChatGPT assistance for visual styling, primarily coloring. ChatGPT was also used to help identify a subset of related work, and the Stanford Agentic Reviewer was used to obtain pre-submission feedback.

AI-assisted tools were also used for programming support, including implementation assistance for the main research artifacts, construction of baselines, and evaluation setup. All AI-assisted outputs were reviewed, edited, and verified by the authors. The authors are fully responsible for the accuracy, originality, and integrity of the manuscript, figures, code, experiments, and conclusions.

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Table 8: Terminology used to connect application requirements, operator contracts, residual disclosure, and contextual integrity.

Term	Meaning	Example
Application output type	Technical representation requested or accepted by an application	person_count, pose_keypoints, sound_event_label
Operator utility capability	Task capability preserved or produced by an operator	occupancy estimation, fall detection support, sound-event recognition
Residual disclosure attribute	Sensitive attribute still exposed after preprocessing	face, identity, speech content, activity, trajectory, gait
CI information type	Contextual-integrity description of what information flows to a recipient	occupancy information, coarse presence, activity information, speech information

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A Implementation Artifacts and Evaluation Details

This appendix provides the implementation and evaluation details that support Section 4. We summarize the structured request format, vocabularies, operator contracts, CI rule library, generated pipeline records, survey construction, downstream evaluators, and dynamic-context compatibility checks. The goal is not to reproduce every JSON object or every generated pipeline, but to give representative examples sufficient to interpret the experiments.

A.1 Background Terminology

The main text uses the following terms to connect application requirements, operator contracts, residual disclosure, and contextual integrity.

A.2 Application Requests

Each downstream application is represented by a structured request. A request records the application identity, the CI context fields supplied by the scenario, admissible sensing sources, and a utility contract that defines acceptable outputs. The same task can be represented in two modes. The *downstream-compatible* mode preserves the interface expected by the fixed downstream evaluation code. The *flexible* mode lists multiple utility-sufficient representations, ordered by interface priority and annotated with representation roles, disclosure tiers, and validation status. This lets the mediator choose a less revealing output when the application contract explicitly declares that representation sufficient.

Table 9: Representative application request families. Downstream-compatible requests preserve the interface of the fixed evaluation model; flexible requests enumerate several representations that the application contract treats as utility-sufficient.

Task	Source modalities	Downstream-compatible outputs	Flexible outputs	Example fields	quality
Visitor presence	Visual cameras	Raw or redacted image frame; raw or redacted video window	Image frame; redacted frame; person detections; person-count time series; binary presence	Latency	bound;
Fall detection	Visual or motion sensing	Pose keypoint sequence for the pose-based fall model	Redacted video for pose extraction; pose keypoints; activity label; guarded fall/safety event	minimum recall	bound;
Domestic sound monitoring	Microphone or smart speaker audio	Raw audio chunk; speech-removed audio chunk	Raw audio; speech-filtered audio; sound-event label; decibel-level window; aggregate sound summary	Latency	bound;
ADL recognition	Synchronized audio-visual sensing	YouHome audio-visual sample or manifest, with optional component-level redaction/filtering	Audio-visual sample; redacted/filter AV sample; pose keypoints; detections; sound-event labels; activity label; aggregate activity summary	minimum precision	recall;
				Latency	bound;
				minimum accuracy	accuracy

Each accepted output cap contains three kinds of information. First, it names a media or semantic representation, such as image/x-row, application/x-pose-keypoints, application/x-detections, or application/x-occupancy-count. Second, it records a schema and an application-facing affordance, such as a frame manifest, pose sequence, sound-event label, or occupancy property. Third, it records CI-relevant annotations, including required information types, required transmission principles, representation role, disclosure tier, and validation status. These annotations allow the planner to distinguish outputs that are mechanically compatible from outputs that are also appropriate for the requested application boundary.

A.3 Vocabulary and Operator Contracts

The implementation uses two related vocabularies. The pipeline vocabulary describes dataflow concepts such as pipelines, operators, ports, caps, affordances, observations, and policy effects. The CI vocabulary describes candidate information flows, including pipeline stage, context, space, sender, subject, recipient, purpose, sensing-data metadata, information type, transmission principle, representation role, and abstraction level. Candidate flows are staged as raw_input, intermediate_representation, output_to_application, or retained_artifact, because the same information may be acceptable internally but unacceptable as an emitted output or retained artifact.

The operator catalog contains symbolic contracts for pre-processing operators. Each contract declares input caps, output caps, utility capabilities, transformation effects, residual-disclosure effects, and CI annotations. Candidate generation uses these declarations before running any operator implementation. The flexible-output catalog contains 28 dataflow operators and three control primitives. The control primitives are not pipeline transformations; they represent policy gating, obligation annotation, and no-compromise resolution.

The runtime implementation mirrors these symbolic operators where executable support is needed. For example, the region-mask operator can use detector annotations or internally run a detector when annotations are absent; the pose extractor defaults to YOLO-style pose keypoints for compatibility with the fall-detection pipeline; speech-content removal

Table 10: Representative vocabulary fields used in generated candidate flows.

Field	Meaning
pipelineStage	Where the flow occurs relative to the preprocessor, such as raw input, intermediate representation, output, or retained artifact.
sensingDataMetadata	Device and modality metadata, such as camera, microphone, image content, video content, audio content, or audio-visual content.
informationType	Data or meaning exposed by the flow, separated into sensor primitives, interpreted observations, and inferred information types.
transmissionPrinciple	Conditions and transformations attached to the flow, such as purpose limitation, data minimization, raw media removed, aggregate only, or speech content removed.
representationRole	The application-boundary role of the output, such as fixed downstream input, semantic primitive, aggregate summary, or final task decision.

can redact transcripts or mute speech-like segments; and occupancy derivation maps detections, motion, or scalar readings into counts or binary occupancy. These implementations are used to build executable pipelines, while the symbolic contracts remain the source of planning-time reasoning.

A.4 CI Rule Library

The CI rule library translates policy and deployment requirements into staged predicates over residual flows. Rules can match context, space, sensing device, content type, recipient, subject, purpose, information type, transmission principle, and pipeline stage. Rule effects include DENY_FLOW, REQUIRE_CONDITION, REQUIRE_TRANSFORMATION, ALLOW_IF, CLASSIFY_AS, and PREFER. The fixed and flexible libraries contain 51 and 54 rules, respectively. The flexible library adds reusable semantic-output and final-task-decision boundary checks; its rules include 21 deny rules, 19 condition requirements, 2 transformation requirements, 6 conditional allowances, 3 classification rules, and 3 preferences.

Each CI rule records provenance metadata: the source name, source type, source natural-language policy statement, and, for external sources, the source URL. The rule sources include platform policies for short-term rentals, venue and institutional surveillance policies, legal or regulatory guidance, professional guidance, research-protocol guidance, and a small

Table 11: Representative operator contracts. The actual catalog contains additional variants and parameters.

Operator	Input/output role	Utility role	Residual effect	CI annotations
op.source	Produces raw video, image, audio, scalar readings, or synchronized AV samples	Supplies raw stream or sensor event to the planner	Initializes residual risk by modality; raw video/audio carry high disclosure for identity, appearance, speech, activity, location, and co-presence	Marks raw sensor primitives and source modality metadata
op.region_select_crop	Media → reduced field-of-view media	Restricts monitoring to an application-relevant spatial region	Reduces off-region context, visible text, and co-presence but preserves information inside the crop	Adds field-of-view minimization and data-minimization terms
op.region_mask_blur	Image/video → redacted image/video	Supports format-preserving downstream compatibility	Reduces targeted visual attributes, such as face, body, person, or background regions, while preserving media shape	Adds redaction and data-minimization terms
op.person_object_detector	Image/video → detections	Supports presence, occupancy, security, and ADL primitives	Removes raw pixels from the emitted output but may retain object location, count, co-presence, and activity cues	Adds interpreted observations such as person or object detected
op.pose_extractor	Image/video → pose keypoints	Supports fall detection, motion analysis, and ADL recognition	Removes texture and raw pixels but may retain posture, gait, body-shape cues, activity, and trajectory	Adds pose-keypoint observation and raw-pixel removal terms
op.speech_content_removal	Audio or transcript → filtered audio, redacted transcript, or intent output	Supports audio applications that do not need speech content	Reduces speech content and may reduce speaker identity, depending on mode	Adds speech-content-removed and data-minimization terms
op.occupancy_deriver	Detections, motion, or readings → count or binary occupancy	Supports occupancy, energy, and visitor-presence applications	Drops appearance and speech attributes; may retain aggregate presence and coarse routines	Adds aggregate or occupancy observation terms
op.multimodal_primitive_fuser	Pose, detections, occupancy, sound, or motion primitives → multimodal primitive record	Supports ADL and safety applications without exposing full AV media	Combines residual risks conservatively across inputs while removing face, speech content, and visible text from the emitted primitive record	Adds semantic-minimization terms

number of created system-design templates for mediator-specific boundaries. Table 12 reports representative externally sourced examples with the specific source URLs used in the constraint files; the full JSON rule artifacts retain the complete natural-language policy statements and additional provenance fields.

Classification and preference rules do not by themselves authorize a flow. Classification rules attach tags that later rules may consume. Preference rules are used for ranking or diagnostics, not for overriding hard constraints. This keeps policy enforcement separate from optimization over otherwise feasible pipelines.

A.5 Generated Pipeline Records, Baselines, and Artifact Inventory

The candidate generator emits structured records for candidate and selected pipelines. A selected pipeline record contains the decision, selected pipeline identifier, matched output cap, final output type and schema, operator sequence, output directory, JSON artifacts, and diagnostic fields such as no-compromise reason and failed-rule summaries. The generated summaries include raw, manual, direct-LLM, full-mediator, and ablation outputs for each context scenario.

The raw baseline forwards source data with no mediation. The manual baseline selects a fixed task/space policy pipeline. The direct-LLM baseline asks an LLM to choose one pipeline directly, without enumerating the candidate set. The full mediator performs symbolic enumeration, utility matching, CI filtering, residual checking, optional probe integration, and least-revealing selection. Ablations remove or alter one component at a time: `utility_only` ignores CI and residual privacy constraints; `no_ci_filter` ignores CI feasibility in final selection; `no_residual_bounds` ignores request residual-disclosure bounds; `no_least_revealing` ranks by utility/cost rather than residual risk after feasibility filtering; `metadata_only` ignores empirical probe residuals; `uniform_risk_weights`

removes context-specific residual-risk weights; `no_staged_flows` collapses staged flows to the output boundary; and `first_feasible` and `latency_first` change the ranking criterion after feasibility filtering.

A.6 Mediator Component Authority

Table 15 summarizes which components can affect the mediator decision. The main point is that candidate generation and final selection are symbolic; optional LLM use is limited to structured norm judgment and cannot override hard rules.

A.7 Mediator Algorithm Reference

The full mediator algorithm is given once in Section 3.6 as Algorithm 1. Earlier drafts duplicated this procedure in the appendix; we omit the duplicate here to keep the appendix focused on implementation artifacts and evaluation details.

A.8 Probe Validation Details

Probe validation is a conservative implementation check over concrete transformed artifacts. The mediator first performs metadata-based selection using operator contracts and CI rules. If probe artifacts and a probe configuration are available for the provisional candidate, the implementation runs privacy probes on those artifacts and attaches the resulting residual evidence to the candidate before reranking.

Probe evidence is combined conservatively. For each residual attribute, the final residual estimate keeps the higher-risk value between the operator metadata and the probe result. Therefore probes can reveal that an implementation leaks more than its contract suggested, but they do not make a candidate appear safer than the symbolic contract. If probe evidence causes the provisional candidate to violate residual bounds or fall below another feasible candidate, the selector can move to the next feasible candidate under the same conservative metadata assumptions. When no probe artifact manifest is

Table 12: Representative CI rules with concrete provenance. The examples use external source URLs recorded in the constraint files and span platform, venue, clinical, workplace, and research settings.

Rule	Source	URL	Effect	Interpretation
airbnb_hidden_recording_devices_ban	Airbnb camera and recording-device policy	https://www.airbnb.com/help/article/3061	DENY_FLOW	Rejects flows from hidden recording devices across raw, intermediate, output, and retained stages.
airbnb_noise_monitor_decibel_only	Airbnb camera and recording-device policy	https://www.airbnb.com/help/article/3061	ALLOW_IF	Allows sound-level outputs only when the flow is decibel-only and speech/audio waveform content is absent.
vrbo_indoor_surveillance_ban	Vrbo surveillance policy	https://help.vrbo.com/articles/What-is-HomeAw https://policies.iu.edu/policies/ps-02-video-electronic-surveillance/index.html	DENY_FLOW	Rejects indoor surveillance-device flows in short-term-rental contexts.
iu_sensitive_workspace_minimization	Indiana University video/electronic policy	https://policies.iu.edu/policies/ps-02-video-electronic-surveillance/index.html	REQUIRE_TRANSFORMATION	Requires field-of-view minimization, screen-content removal, and data minimization for sensitive workspace camera flows.
hhs_blur_not_sufficient_without_authorization	HHS HIPAA media and film-crew guidance	https://www.hhs.gov/sites/default/files/guidance-on-media-and-film-crews-access-to-phi.pdf	DENY_FLOW unless authorized	Treats transformation alone as insufficient for public/media release without explicit authorization.
uoregon_bystander_and_nonface_identifiers	University of Oregon human-subjects audio/video/photo guidance	https://research.uoregon.edu/manage/integrity-compliance/human-subjects-research/guidance-library/audio-video-photography	REQUIRE_TRANSFORMATION	Requires research recordings to address non-face identifiers and bystanders through identity removal, field-of-view minimization, speech-content removal, or consent/exclusion of non-consenting people.

Table 13: Representative generated pipeline decisions. These examples are selected to show different output representations and decision outcomes rather than to enumerate every scenario.

Mode / scenario	Decision	Matched output	Final output	Operator sequence
Fixed S001	Select	Visitor image-frame interface	image/x-redacted; redacted frame	op.source->op.sample->op.region_select_crop->op.region_mask_blur->op.route_publish
Fixed S007	Select	LE2I pose-sequence interface	application/x-pose-keypoints; pose keypoints	op.source->op.pose_extractor->op.route_publish
Fixed S013	Select	YouHome redacted/filter AV interface	application/x-youhome-av-sample; AV sample/manifest	op.source->op.av_visual_redact->op.av_audio_speech_filter->op.route_publish
Fixed S019	Select	CHiME-Home removed audio	audio/x-filtered; speech-removed waveform	op.source->op.sample->op.trigger_gate->op.speech_content_removal->op.route_publish
Fixed S002	No compromise	None	None	No candidate passed CI evaluation; diagnostics record the failed policy rules.
Flexible S003	Select	Visitor person detections	application/x-detections; object detections	op.source->op.person_object_detector->op.route_publish
Flexible S004	Select	Visitor count time series	application/x-occupancy-count; occupancy count	op.source->op.region_mask_blur->op.silhouette_extractor->op.aggregate_generalize->op.schema_adapter->op.route_publish
Flexible S024	Select	CHiME-Home removed audio	audio/x-filtered; speech-removed waveform	op.source->op.sample->op.trigger_gate->op.speech_content_removal->op.route_publish

Table 14: Implementation artifact inventory.

Artifact class	Role
Application request JSON files	Define task capability, source constraints, quality fields, accepted output caps, and rejected output caps.
Pipeline vocabulary JSON	Defines caps, dataflow concepts, operator interface fields, affordances, and representation metadata.
CI vocabulary JSON	Defines staged flow fields, information types, transmission principles, and rule matching vocabulary.
Operator contracts JSON	Defines symbolic operator effects used during candidate generation and residual propagation.
CI constraints JSON	Defines hard constraints, conditional allowances, transformation requirements, classifications, and preferences.
Runtime operator implementation	Provides executable versions of selected preprocessing operators for generated pipelines.
Generated summary JSON	Records selected pipelines, no-compromise decisions, baselines, ablations, matched output caps, and operator sequences by context scenario.

Table 15: Core components and decision authority in the implemented mediator.

Component	LLM?	Role
Structured request loading	Optional upstream	Load an application request containing a utility contract and CI context. Natural-language parsing is outside the evaluated mediator.
Candidate generation	No	Generate candidate preprocessing pipelines from a fixed typed operator-effect catalog.
Symbolic utility check	No	Check accepted output representations and declared task capabilities against the utility contract.
Residual propagation	No	Compose operator effects to estimate residual disclosure.
Layered CI evaluation	Optional bounded judgment	Apply CI rule-library constraints over residual CI flows. Optional LLM use is restricted to structured norm judgment and cannot override hard rules.
Probe validation	No	Optionally combine metadata residuals with empirical probe residuals for concrete transformed artifacts.
Final selection	No	Select the least-revealing feasible pipeline or reject the application request.

supplied, final selection uses metadata residuals alone. Thus, probes reduce misplaced trust in implementations, but they are not claimed to be exhaustive privacy proofs over the entire candidate space.

B Additional Evaluation Details

B.1 Detailed Main-Body Evaluation Tables

This subsection collects detailed tables moved from the main body to keep Section 4 focused on the coverage-aware evaluation story.

B.2 Dataset Details, Downstream Evaluators, and Flexible Adapters

For visitor presence detection, we use the ChokePoint dataset [50] as a proxy for entryway or portal monitoring. The downstream detector produces frame-level person counts and a binary person-present output. We evaluate the binary output against dataset annotations using precision, recall, and F1/F2, and use count error only as a secondary diagnostic because the application contract requires presence rather than exact identity or tracking.

For fall detection, we use the Le2i Fall dataset [12] as a proxy for safety monitoring in homes, care settings, and clinical spaces. The downstream application operates on pose keypoints, either supplied directly by a preprocessing pipeline or extracted from video before classification. The fall classifier predicts fall versus non-fall labels over pose sequences/windows, and we report macro-F1 over labeled fall/non-fall examples as the primary utility metric.

For ADL recognition, we use the YouHome ADL dataset [42] to represent routine activity monitoring and personalization in smart-home or care contexts. The downstream model consumes synchronized visual frames and audio segments and predicts an ADL class. We report macro-F1 and accuracy over held-out ADL labels. The evaluator also records missing-image and missing-audio counts as validity checks for materialized preprocessing outputs.

For domestic sound monitoring, we use the CHiME-Home audio dataset [23] to represent smart-space monitoring of household sounds, alarms, non-speech events, and voice-related activity. The downstream model is a multi-label audio tagger. We use macro-F1 as the primary utility metric because the task is multi-label and label frequencies are uneven; micro-F1, sample-F1, and per-label predictions are retained as secondary diagnostics.

In the flexible-input setting, selected outputs are evaluated at the interface declared by the application contract. This applies uniformly to MANUAL, DIRECT-LLM, *privoMediator*, and the ablations. Outputs that preserve the original downstream interface are evaluated by the fixed task evaluator. Accepted semantic outputs, such as detections, occupancy counts, pose keypoints, sound-event labels, or final task-event records, are evaluated through the corresponding flexible-output adapter. Raw forwarding is unchanged between fixed-input and flexible-input settings, so raw metrics are reused as the reference denominator.

B.3 Survey Instrument, Interface, and Additional Results

The survey presented generated information flows as short participant-facing vignettes rather than as raw CI tuples. Each page asked participants to rate a complete situation from the perspective of the person the data is about. The page explicitly instructed participants to judge appropriateness rather than legality or whether a space owner was allowed to use the device. To reduce ambiguity, the page also stated that only the listed data was saved or shared and that only the named recipient could access it. When adjacent questions were variants of the same scenario, rows that differed from the prior question were visually marked as changed, while unchanged rows still applied.

Participants were recruited through Prolific and assigned to either the fixed-input arm or the flexible-input supplemental arm. Each participant rated a randomized subset of scenario-output items. Items were shown without method labels. The survey used explicit attention checks and scenario-parameter comprehension checks. The completed-session dataset contains 82 respondents; one participant did not complete the survey and was excluded. Participants answered all 164 attention checks correctly, and one of 82 comprehension checks failed.

The survey stimuli were generated from a context-only scenario pool and a participant-facing glossary. The context pool contained 33 deduplicated scenarios spanning four tasks. These context shells contained the CI context fields—context, space, sender, subject, recipient, purpose, and transmission principle—but left the shared-output fields blank until populated by generated system or baseline outputs. The readable glossary mapped machine terms such as `short_term_rental`, `fall_detection`, `recipient`, and `speech_content_removed` into plain-language survey text.

Because multiple ratings come from the same participant and each scenario-output item can receive multiple ratings, item-level ratings are not independent. The main paper therefore reports descriptive repeated-measures summaries rather than treating each rating as an independent observation. A fuller inferential analysis could use clustered standard errors or a mixed-effects model with participant and item random effects, for example:

$$\text{rating}_{p,i} \sim \text{output class} + (1|p) + (1|i).$$

B.4 Dynamic-Context Utility Compatibility

Table 29 reports utility for the modality-fixed dynamic-context evaluation. These results should be interpreted as a compatibility check under dynamic switching rather than a replacement for the full utility evaluation in Section 4.3: non-output decisions are not evaluated by the downstream model, and the run uses a bounded selected-output sample. In modality-fixed applications, the downstream model constrains the mediator to remain compatible with a fixed input interface. This limits the privacy-utility tradeoff because privacy-preserving

Table 16: Contract-derived residual-disclosure accounting for selected outputs. Lower values indicate less exposure under the operator-effect metadata; this table is descriptive and not a formal leakage measurement.

Run	Method	Sel.	Raw-like	Mean risk	High attr	Med/high attr	ID $\geq$ med	Face $\geq$ med	Speech $\geq$ med
Fixed	RAW	33	25	27.5	7.6	9.8	72.7%	72.7%	51.5%
Fixed	MANUAL	33	0	18.0	3.2	5.9	48.5%	0.0%	15.2%
Fixed	<i>privMediator</i>	18	0	17.4	3.1	5.9	44.4%	0.0%	0.0%
Flexible	RAW	33	25	26.3	7.1	9.4	72.7%	72.7%	51.5%
Flexible	MANUAL	33	0	16.3	2.9	5.3	15.2%	0.0%	15.2%
Flexible	<i>privMediator</i>	21	7	14.4	3.0	4.9	19.0%	19.0%	14.3%

Table 17: Probe-validation examples for operator-effect assumptions. Probe evidence is used as an implementation check and can conservatively increase, but not decrease, residual-disclosure estimates.

Contract assumption	Probe example	Residual attributes checked	Interpretation
Face/visual redaction suppresses identifying visual details	Face detector and OCR after image/video blur or AV visual redaction	face, identity, visible text	A remaining face or text detection raises the corresponding residual attribute; absence of a detection is only a sanity check, not a guarantee.
Speech filtering removes intelligible speech content	ASR after speech-content removal or filtered-audio transforms	speech content	Recognized words after filtering indicate residual speech content and raise the candidate risk.
Audio filtering may leave speaker or co-presence cues	Diarization or speaker-recognition-style probes after speech filtering	speaker identity, co-presence	Remaining separable speakers or speaker-like embeddings are treated as residual speaker/co-presence evidence.
Pose extraction removes pixels but preserves body dynamics	Person/pose consistency checks on extracted keypoints	body shape, gait, activity, trajectory	Keypoints support task utility but still expose motion and activity structure; the contract records these residuals rather than treating pose as anonymous.
Semantic summaries reduce raw-media exposure	Person/object or event-output sanity checks after detection/count/event operators	aggregate presence, activity, location	Summaries are checked for schema and task signal while retaining only the residual attributes declared by the operator contract.

Table 19: Fixed-input downstream utility. Utility is conditional on selected outputs and is not coverage-adjusted.

Task	Method	Sel./rej.	Metric	Ret.
Visitor	RAW	8/0	0.733	1.000
Visitor	MANUAL	8/0	0.713	0.973
Visitor	DIRECT-LLM	5/3	0.713	0.973
Visitor	<i>privMediator</i>	4/4	0.713	0.973
Fall	RAW	8/0	0.941	1.000
Fall	MANUAL	8/0	0.941	1.000
Fall	DIRECT-LLM	8/0	0.941	1.000
Fall	<i>privMediator</i>	6/2	0.941	1.000
ADL	RAW	8/0	0.664	1.000
ADL	MANUAL	8/0	0.678	1.021
ADL	DIRECT-LLM	7/1	0.678	1.021
ADL	<i>privMediator</i>	4/4	0.678	1.021
Sound	RAW	9/0	0.646	1.000
Sound	MANUAL	9/0	0.559	0.866
Sound	DIRECT-LLM	8/1	0.559	0.866
Sound	<i>privMediator</i>	4/5	0.559	0.866

Table 20: Flexible-input downstream utility. Raw rows reuse fixed-input metrics; non-raw outputs use fixed evaluators or flexible adapters.

Task	Method	Sel./rej.	Metric	Ret.
Visitor	RAW	8/0	0.733	1.000
Visitor	MANUAL	8/0	0.774	1.056
Visitor	DIRECT-LLM	8/0	0.479	0.653
Visitor	<i>privMediator</i>	7/1	0.629	0.858
Fall	RAW	8/0	0.941	1.000
Fall	MANUAL	8/0	0.720	0.765
Fall	DIRECT-LLM	8/0	0.702	0.746
Fall	<i>privMediator</i>	6/2	0.757	0.804
ADL	RAW	8/0	0.664	1.000
ADL	MANUAL	8/0	0.678	1.021
ADL	DIRECT-LLM	8/0	0.678	1.021
ADL	<i>privMediator</i>	4/4	0.678	1.021
Sound	RAW	9/0	0.646	1.000
Sound	MANUAL	9/0	0.559	0.866
Sound	DIRECT-LLM	9/0	0.559	0.866
Sound	<i>privMediator</i>	4/5	0.624	0.966

Table 18: Diagnostics for rejected *privMediator* requests.

Dominant diagnostic	Cases	Example scenarios	Common rule/condition
Hard denial: hidden or undisclosed monitoring	4	S002, S009, S016, S031	hidden short-term-rental devices; indoor camera/audio bans; disclosure or disable requirements
Hard denial: private or sensitive spaces	4	S010, S022, S030, S032	private-space camera bans; missing required notice/consent for recording
CI conflict: routine surveillance or workplace monitoring	2	S004, S015	routine surveillance bans; sensitive-workspace minimization
CI conflict: research or bystander protocol limits	1	S017	explicit consent; bystander/non-face identifier limits; protocol metadata
CI conflict: outdoor or location policy conditions	2	S003, S034	disclosed coverage/location requirements; outdoor surveillance limited to security purposes
CI conflict: audio surveillance constraints	2	S021, S033	indoor audio limits; smart-device disclosure or disable requirements

transformations can satisfy the interface while still removing signal that the downstream model relied on.

The zeros in Table 29 should not be read as contradicting the main utility evaluation. They arise in a bounded modality-fixed compatibility run where selected transformations satisfy an interface but remove signal used by the particular downstream model on the selected sample. The main utility evaluation provides the broader task-level utility results over evaluated selected outputs. The dynamic experiment is

Table 21: Method-linked appropriateness ratings over deduplicated scenario-output pairs. Selection counts are fixed/flexible; confidence intervals are participant-cluster bootstrap intervals for mean ratings.

Method	Fixed sel.	Flex sel.	Rows	Mean	95% CI	≥ 4	Fixed CI-conflict rel.	Interpretation
RAW	33/0	33/0	446	2.83	[2.64, 3.02]	40.4%	15	raw media is the least appropriate selected-output condition
MANUAL	33/0	33/0	327	3.43	[3.24, 3.63]	59.3%	15	transformations improve ratings but do not add contextual selectivity
DIRECT-LLM	28/5	33/0	371	3.56	[3.37, 3.74]	60.9%	11	often selects transformed outputs, but still releases in conflict contexts
<i>privMediator</i>	18/15	21/12	215	3.72	[3.51, 3.90]	66.0%	0	combines appropriate selected outputs with configured-rule refusal

Table 22: Ablation results over 33 scenarios. Utility and survey scores are computed over selected outputs.

Variant	Fixed sel.	Flex sel.	Fixed util.	Flex util.	Mean rating	Approp.	Interpretation
Full <i>privMediator</i>	18/15	21/12	0.965	0.912	3.72	66.0%	balances utility, CI feasibility, and least-revealing selection
UTILITY-ONLY	33/0	33/0	1.000	0.951	2.97	44.2%	preserves utility by releasing outputs even in CI-conflict contexts
NO CI FILTER	33/0	33/0	0.965	0.950	3.45	59.5%	privacy transforms help, but hard-rule-violating outputs remain
NO LEAST-REVEALING	18/15	21/12	0.989	0.989	3.31	52.5%	keeps refusals but often chooses more revealing feasible outputs
LATENCY-FIRST	18/15	21/12	0.989	0.989	3.31	52.5%	prioritizes faster pipelines over least disclosure in this run

Note. Utility retention is averaged over evaluated selected outputs. For NO LEAST-REVEALING and LATENCY-FIRST, utility averages exclude fall detection because no evaluated fall outputs were present.

Table 23: Downstream utility evaluators and flexible-output adapters. Fixed evaluators consume the representations expected by the original task pipelines. Flexible adapters evaluate semantic outputs that the app contract accepts as utility-sufficient.

Task	Fixed downstream evaluator	Compatible fixed input	Flexible semantic adapter role
Visitor presence detection	ChokePoint YOLO person-presence evaluator	Image frames, frame video/frame manifests	Converts detections, counts, or binary presence into the prediction CSV format used by the ChokePoint metric code, without rerunning YOLO.
Fall detection	LE2I pose-GRU fall classifier	Pose keypoint windows or pose sequences extracted from video	Converts pose keypoints or accepted fall/safety-event outputs into fall/nonfall predictions and computes binary metrics.
ADL recognition	YouHome audio-visual ADL classifier	Synchronized image frames and audio waveform samples, with optional fixed-interface redaction/filtering	Converts accepted ADL semantic labels into multiclass predictions; these are proxy metrics unless the app contract declares that semantic label boundary as final.
Domestic sound monitoring	CHiME-Home audio tagging evaluator	Audio waveform chunks loaded into log-mel/CNN or AST-style features	Converts sound-event labels or decibel summaries into CHiME-style multilabel predictions; decibel-only outputs are treated as coarse noise-monitoring proxies rather than full event-tagging equivalents.

Table 26: Fixed-input survey ratings by output class. Less revealing outputs are generally rated more appropriate than raw audio, raw AV, or raw media.

Output class	Items	Ratings	Mean	≥ 4
Derived pose	16	143	3.72	66.4%
Redacted media	8	89	3.56	60.7%
Redacted+filtered AV	15	143	3.51	59.4%
Filtered audio	12	125	3.34	59.2%
Redacted AV	5	49	3.20	53.1%
Raw media	24	311	3.12	50.2%
Raw AV	8	111	2.91	42.3%
Raw audio	9	109	2.39	24.8%

Table 27: Flexible-input policy selectivity. Flexible contracts allow more abstract outputs, but selection still depends on whether any accepted representation satisfies the configured CI rules.

Method	Sel.	Rej.	CI-conflict rel.	Hard-rule viol.	Interpretation
RAW	33	0	12	5	always forwards raw streams
MANUAL	33	0	12	5	always selects a task policy
DIRECT-LLM	33	0	12	5	selects an output in every flexible scenario
<i>privMediator</i>	21	12	0	0	rejects when no CI-feasible flexible output exists

Table 28: Dynamic context-switching behavior and latency across 17 context phases. Unsafe outputs are releases that violate configured hard rules.

Method	Output	Review/reject	Unsafe	Median	P95
RAW	17/17	0/17	4/17	4.3 ms	5.2 ms
MANUAL	17/17	0/17	4/17	2.15 s	4.33 s
DIRECT-LLM	15/17	2/17	2/17	2.89 s	4.54 s
<i>privMediator</i>	12/17	5/17	0/17	2.28 s	4.35 s

Table 29: Utility of selected outputs in the dynamic-context, modality-fixed evaluation. Metrics are task-specific: macro-F1 for ADL, accuracy for visitor presence, video-level macro-F1 for fall detection, and micro-F1 for domestic sound monitoring. Missing phases for *privMediator* correspond to rejected requests rather than failed downstream execution. Raw, Man., D-LLM, and Full denote RAW, MANUAL, DIRECT-LLM, and *privMediator*, respectively.

Task	Metric	Raw	Man.	D-LLM	Full
ADL recognition	Macro-F1	1.00	0.00	0.00	0.00
Visitor presence	Accuracy	1.00	1.00	1.00	1.00
Fall detection	Video F1	1.00	0.00	0.00	0.00
Domestic sound	Micro-F1	0.67	0.67	0.67	0.67

Table 30: Median *privMediator* context-switch overhead by mediator stage in the dynamic-context experiment.

Stage	Median time
Request loading	0.1 ms
Candidate generation	2.15 s
CI evaluation	34.1 ms
Preliminary selection	6.9 ms
Probe stage	< 0.1 ms
Final selection	6.9 ms
Total mediator decision	2.28 s

Table 24: Flexible-input survey ratings by output class. Participants generally rate event records and summaries as more appropriate than lower-level room-occupied indicators.

Output class	Items	Ratings	Mean	≥ 4
Fused event record	6	76	4.04	84.2%
Activity summary	4	54	3.65	63.0%
Sound-event summary	4	53	3.57	66.0%
Person detections	6	73	3.53	57.5%
Occupancy count	7	99	3.43	62.6%
Room occupied	3	41	2.61	31.7%

Table 25: Survey data summary. Ratings are completed-session ratings after excluding one incomplete participant.

Quantity	Fixed	Flexible	Total
Completed respondents	60	22	82
Survey items	97	30	127
Ratings	1080	396	1476
Ratings per item	11.13	13.20	11.62
Mean rating	3.22	3.53	3.30
Median rating	4	4	4
Ratings ≥ 4	52.2%	63.1%	55.1%
Optional comments	257	75	332
Attention checks passed	120/120	44/44	164/164
Comprehension checks passed	59/60	22/22	81/82

included to test context-switch behavior and unsafe-output avoidance, not to re-estimate full task utility.

Scenario details

What is the scenario?	A monitoring device in the main office floor of a workplace listens for sound events such as alarms, footsteps, or glass breaking to support safety by detecting concerning sounds, such as alarms, breaking glass, yelling, or other urgent audio cues.
What is your role in this scenario?	For this scenario, imagine the data is about you as an employee on the main office floor of this workplace.
What data would be shared?	The shared data is a sound-event summary report: a weekly chart or written summary showing broad counts or percentages for noise types such as alarms, footsteps, loud sounds, and quiet time. The original audio recording is never saved or shared, and it does not include words from conversations. The report uses broad weekly counts or percentages rather than a live transcript or minute-by-minute audio log. The original recording and spoken words are not shared. Example: a weekly chart showing: alarms triggered: 2 times; footstep noise: 15% of the day; quiet hours: 80% of the day.
Who controls the monitoring device?	The device is a microphone controlled by the organization responsible for the workplace.
Who receives or uses the shared data?	Authorized staff receive the data for sound monitoring in the workplace. Only this named receiver or listed app/service can access or use the shared data; do not assume device makers, cloud providers, landlords, managers, or other unlisted parties can access it.
What notice or permission is given?	You are told that monitoring or data sharing is happening, such as through a sign, notice, or setup screen. Example: for example, a sign, notice, or setup screen tells people monitoring is happening

Appropriateness rating

In your personal judgment, how appropriate is it for the monitoring device to share the listed data in this situation? Answer from the role shown on the page. Here, appropriate means the sharing seems acceptable and justified for this specific setting, person, receiver, purpose, rule or condition, and type of data shared.

○

1

Completely inappropriate

○

2

Somewhat inappropriate

○

3

Neutral / unsure

○

4

Somewhat appropriate

○

5

Completely appropriate

Figure 4: Example survey interface. The interface shows the scenario description, the participant’s role, the data that would be shared, the device controller, the recipient, and the notice or processing condition, followed by a five-point appropriateness rating, confidence prompt, and optional explanation.